November 2, 2026

Cox quotients, torus torsors, and line bundles

These notes follow the lecture and board images, using the notation of Chapter 6, *Cox Coordinates and Quotient Constructions*. The main points are:

- Affine product descriptions can glue to a nontrivial torus torsor.
- A character of the Cox group specifies a line bundle; its weight is one additional column in the action matrix.
- The irrelevant locus removes exactly the coordinate strata where the torus map cannot extend to a toric morphism.
- Quotients are constructed from invariant regular functions. Smooth, simplicial, and nonsimplicial fans give different orbit behavior.

1 Notation and the Cox construction

Work over $\mathbb{C}$. Let $N \cong \mathbb{Z}^d$ be the lattice of *one-parameter subgroups*, $M = \mathrm{Hom}(N, \mathbb{Z})$ its character lattice, and Σ a fan whose rays span $N_{\mathbb{R}}$. Put $r = |\Sigma(1)|$, and write u_ρ for a primitive ray generator. The two lattice maps are

$$\beta : \mathbb{Z}^r \longrightarrow N, \quad e_\rho \longmapsto u_\rho,$$
$$\beta^* : M \longrightarrow \mathbb{Z}^r, \quad m \longmapsto (\langle m, u_\rho \rangle)_\rho.$$

Remark 1 (The lattice convention on the board). The map sending standard basis vectors to ray generators is a map of N-lattices. Some board labels call these spaces M. Here we use Chapter 6's convention: cones lie in $N_{\mathbb{R}}$, while characters and semigroup coordinate rings use M.

Recall the divisor-class sequence and its diagonal group:

$$0 \longrightarrow M \xrightarrow{\beta^*} \mathbb{Z}^r \xrightarrow{\deg} \mathrm{Cl}(X_\Sigma) \longrightarrow 0, \qquad G = \mathrm{Hom}(\mathrm{Cl}(X_\Sigma), \mathbb{G}_m).$$

The Cox ring, irrelevant ideal, and Cox open set are

$$S = \mathbb{C}[x_\rho : \rho \in \Sigma(1)], \quad \deg(x_\rho) = [D_\rho],$$
$$B(\Sigma) = \left\langle x^{\widehat{\sigma}} : \sigma \in \Sigma_{\max} \right\rangle, \qquad x^{\widehat{\sigma}} = \prod_{\rho \not\in \sigma(1)} x_\rho,$$
$$Z(\Sigma) = V(B(\Sigma)), \qquad \widehat{X}_\Sigma = \mathbb{A}^r \setminus Z(\Sigma).$$

The quotient morphism is

$$\pi : \widehat{X}_\Sigma \longrightarrow X_\Sigma, \qquad X_\Sigma \cong \widehat{X}_\Sigma \,/\!/\, G.$$

The symbol $Z(\Sigma)$ denotes a zero set, whereas $\mathbb{Z}$ denotes the integers. We remove $V(B)$; we do not replace S by S/B.

2 Local products need not give a global product

Suppose a surjective lattice map $\beta : \widehat{N} \twoheadrightarrow N$, with kernel K, identifies a cone $\widehat{\sigma}$ with σ and identifies their saturated span lattices integrally. An adapted splitting gives

$$(\widehat{N}, \widehat{\sigma}) \cong (N \oplus K, \sigma \times \{0\}), \qquad U_{\widehat{\sigma}} \cong U_\sigma \times T_K,$$

where $T_K \cong (\mathbb{G}_m)^{\operatorname{rank} K}$. This follows by taking dual semigroup algebras: the kernel directions contribute invertible Laurent variables. The splitting is a choice, so the product is not canonical.

For a smooth Cox fan, with $\beta : \mathbb{Z}^r \to N$ surjective, this yields

$$\pi^{-1}(U_\sigma) \cong U_\sigma \times (\mathbb{G}_m)^{r-d}.$$

Different chartwise splittings can have nonconstant transition functions. Consequently, the global conclusion is a *torus torsor over* X_Σ, not necessarily a product.

Definition 1 (Torus torsor in the setting of the lecture)**.** A Zariski locally trivial T-torsor $P \to X$ is a space with a T-action and an open cover $\{U_i\}$ such that

$$P|_{U_i} \cong U_i \times T$$

equivariantly, with T acting on itself by multiplication. On each fiber the action is free and transitive. The torsor is globally trivial exactly when it admits a global section.

Remark 2 (Hypothesis behind the torus-bundle statement)*.* For the smooth torus-bundle statements we assume that β is surjective on lattices. This holds if the smooth fan has a full-dimensional cone, as in the lecture examples. It implies that $\mathrm{Cl}(X)$ is free and $G \cong (\mathbb{G}_m)^{r-d}$. Smoothness alone, for arbitrary noncomplete fans, does not force the global class group to be torsion free. Without the additional hypothesis the smooth Cox quotient is a torsor under a diagonalizable group, locally trivial in the appropriate *étale* topology; it need not be a Zariski torus torsor.

2.1 The first example already has nontrivial gluing

For $\mathbb{P}^1$, the Cox map is

$$\pi : \mathbb{A}^2 \setminus \{0\} \longrightarrow \mathbb{P}^1, \qquad (x, y) \longmapsto [x : y].$$

Its fibers are scalar orbits, each isomorphic to $\mathbb{G}_m$. The map

$$(x, y) \longmapsto \big([x : y], (x, y)\big)$$

identifies its source with the tautological line bundle with its zero section removed:

$$\boxed{\mathbb{A}^2 \setminus \{0\} \cong \operatorname{Tot} \mathcal{O}_{\mathbb{P}^1}(-1) \setminus 0_{\mathbb{P}^1}.}$$

Here the fiber of $\mathcal{O}(-1)$ over a line $\ell \subset \mathbb{C}^2$ is the line ℓ itself.

On $U_0 = \{x \neq 0\}$, use base coordinate $u = y/x$ and fiber coordinate $a = x$. On $U_1 = \{y \neq 0\}$, use $v = x/y$ and $b = y$. The overlap transformation is

$$\boxed{v = u^{-1}, \qquad b = ua.}$$

The fiber transition depends on the base coordinate. The torsor is not $\mathbb{P}^1 \times \mathbb{G}_m$ over $\mathbb{P}^1$: a section would give a nowhere-zero regular section of $\mathcal{O}(-1)$, whereas $H^0(\mathbb{P}^1, \mathcal{O}(-1)) = 0$.

$$\begin{array}{ccc} \pi^{-1}(U_0) \cong \mathbb{A}^1_u \times \mathbb{G}_m & \xrightarrow{(u,a)\mapsto(u^{-1},ua)} & \pi^{-1}(U_1) \cong \mathbb{A}^1_v \times \mathbb{G}_m \\ \Big\downarrow{\scriptstyle \pi} & & \Big\downarrow{\scriptstyle \pi} \\ U_0 \cong \mathbb{A}^1_u & \xrightarrow[u\mapsto u^{-1}]{} & U_1 \cong \mathbb{A}^1_v \end{array}$$

Diagram 1: The dashed coordinate changes are defined on the overlaps, where $u, v \neq 0$. Each chart is a product, but the fiber gluing $b = ua$ gives the nontrivial punctured tautological bundle.

3 Line bundles from one additional Cox coordinate

Continue in the smooth setting, where the Cox quotient $\pi : \widehat{X} \to X_\Sigma$ is the torsor discussed above. For a divisor class $\alpha \in \mathrm{Pic}(X_\Sigma) = \mathrm{Cl}(X_\Sigma)$, evaluation defines a character

$$\chi_\alpha : G \longrightarrow \mathbb{G}_m, \qquad \chi_\alpha(g) = g(\alpha).$$

Add one affine coordinate z, and let

$$g \cdot (x, z) = \big(g \cdot x, \chi_\alpha(g) z\big).$$

Then

$$\boxed{\mathrm{Tot}\, \mathcal{O}_{X_\Sigma}(D) \cong (\widehat{X} \times \mathbb{A}^1)/G, \qquad [D] = \alpha.}$$

The projection to X_Σ sends $[x, z]$ to $\pi(x)$, and $[x, 0]$ defines the zero section. Thus the quotient is the *total space* of the line bundle, rather than its base.

$$\begin{array}{ccc} \widehat{X} \times \mathbb{A}^1 & \xrightarrow{/G} & \mathrm{Tot}\, \mathcal{O}(D) \\ \Big\downarrow{\scriptstyle \mathrm{pr}_1} & & \Big\downarrow{\scriptstyle p} \\ \widehat{X} & \xrightarrow[\pi]{} & X_\Sigma \end{array}$$

Diagram 2: A character of the Cox group determines the action on the extra coordinate. The quotient of a trivial line upstairs can be a nontrivial line bundle downstairs.

3.1 Local coordinates verify the construction

Choose an invariant representative $D = \sum_\rho a_\rho D_\rho$. For a maximal cone σ, let m_σ be its Cartier datum:

$$\langle m_\sigma, u_\rho \rangle = -a_\rho \qquad (\rho \subseteq \sigma).$$

On the Cox chart $\widehat{U}_\sigma = D(\widehat{x^\sigma})$, set

$$q_\sigma = \prod_\rho x_\rho^{a_\rho + \langle m_\sigma, u_\rho \rangle}.$$

Its exponents on rays inside σ are zero; coordinates on the other rays are invertible on this chart. Thus q_σ is an invertible Laurent monomial of degree α, with $q_\sigma(gx) = \chi_\alpha(g) q_\sigma(x)$. Consequently,

$$w_\sigma = \frac{z}{q_\sigma}$$

is invariant, and the local quotient is

$$\operatorname{Spec}(\mathcal{O}(U_\sigma)[w_\sigma]) \cong U_\sigma \times \mathbb{A}^1.$$

On overlaps,

$$w_\tau = \frac{q_\sigma}{q_\tau} w_\sigma = \chi^{m_\sigma - m_\tau} w_\sigma.$$

These are the fiber-coordinate transitions of $\mathcal{O}(D)$: its local frames satisfy $e_\sigma = \chi^{m_\sigma - m_\tau} e_\tau$. This verifies the bundle and fixes the sign of the extra weight.

3.2 The extra column in the weight matrix

Suppose $G \cong \mathbb{G}_m^k$, and let $Q = (q_{i\rho})$ be the $k \times r$ Cox weight matrix. Its ρ-th column records $[D_\rho]$ in the chosen basis of $\mathrm{Cl}(X_\Sigma)$. Thus

$$(t_1, \dots, t_k) \cdot x_\rho = \left(\prod_{i=1}^{k} t_i^{q_{i\rho}} \right) x_\rho.$$

Writing $a = (a_\rho)$, the class of D is the column Qa. The action on the new coordinate is

$$z \longmapsto \left(\prod_{i=1}^{k} t_i^{(Qa)_i} \right) z, \qquad \boxed{Q_L = (Q \mid Qa).}$$

This is the lecture's rule: add the same linear combination of the old columns as the divisor's linear combination of boundary classes. The admissible open set remains $\widehat{X} \times \mathbb{A}^1$; the irrelevant ideal extends from S to $S[z]$.

Example 1 (Calibrating on $\mathbb{P}^1$)**.** For $\mathcal{O}_{\mathbb{P}^1}(n)$, use

$$\lambda \cdot (x_0, x_1, z) = (\lambda x_0, \lambda x_1, \lambda^n z), \qquad Q_L = (1 \;\; 1 \;\; n).$$

On $x_i \neq 0$, the invariant fiber coordinate is $w_i = z/x_i^n$. Hence

$$w_1 = \left(\frac{x_0}{x_1} \right)^n w_0,$$

which is the transition for $\mathcal{O}(n)$. The exponent is $+n$ for the simultaneous action used here. A homogeneous Cox polynomial f of degree $[D]$ similarly gives a section $\pi(x) \mapsto [x, f(x)]$, since f transforms like z.

4 Fan of the total space

The quotient has an explicit fan in $N_L = N \oplus \mathbb{Z}$. For each base ray and for the zero section, take

$$\widetilde{u}_\rho = (u_\rho, -a_\rho), \qquad u_z = (0, 1).$$

For every $\sigma \in \Sigma$, form

$$\widetilde{\sigma} = \mathrm{Cone}(\{\widetilde{u}_\rho : \rho \subseteq \sigma\} \cup \{u_z\}),$$

and include all faces. These cones make the fan of $\operatorname{Tot} \mathcal{O}(D)$. The projection $(v, t) \mapsto v$ induces its bundle map; the vertical ray u_z corresponds to the zero-section divisor.

If the $r \times d$ matrix A has rows u_ρ, the augmented ray matrix is

$$\boxed{A_L = \begin{pmatrix} A & -a \\ 0 & 1 \end{pmatrix}.}$$

It also gives the character-lattice map $M \oplus \mathbb{Z} \to \mathbb{Z}^{r+1}$. The two matrix recipes are compatible:

$$Q_L A_L = (Q \mid Qa) \begin{pmatrix} A & -a \\ 0 & 1 \end{pmatrix} = 0,$$

because $QA = 0$.

4.1 Local products and changing the divisor representative

On σ, the lifted base rays lie on the graph $t = \langle m_\sigma, v \rangle$. Thus

$$\widetilde{\sigma} = \{(v,t) : v \in \sigma,\ t \geq \langle m_\sigma, v \rangle\}.$$

The integral shear $(v,t) \mapsto (v, t - \langle m_\sigma, v \rangle)$ identifies this cone with $\sigma \times \mathbb{R}_{\geq 0}$, proving the local product $p^{-1}(U_\sigma) \cong U_\sigma \times \mathbb{A}^1$. The character $(-m_\sigma, 1)$ supplies its fiber coordinate.

If $D' = D + \operatorname{div}(\chi^m)$, then $a'_\rho = a_\rho + \langle m, u_\rho \rangle$. The global lattice automorphism

$$(v,t) \longmapsto (v, t - \langle m, v \rangle)$$

sends the fan for D to that for D', fixes $(0,1)$, and preserves projection to the base. Also $Qa' = Qa$, so the extra Cox weight depends only on the line-bundle class.

Example 2 (Two representatives of $\mathcal{O}_{\mathbb{P}^1}(n)$)**.** Let the base rays be $+1, -1$, with divisors D_0, D_∞. For $D = nD_\infty$, the lifted rays are

$$(1,0), \qquad (-1,-n), \qquad (0,1).$$

The maximal cones pair the vertical ray with each of the other two. For $D' = nD_0$, the rays instead are

$$(1,-n), \qquad (-1,0), \qquad (0,1).$$

The shear $(v,t) \mapsto (v, t - nv)$ identifies these fans, reflecting $nD_0 \sim nD_\infty$. A change of lattice coordinates must act on the whole fan, rather than on an individual ray.

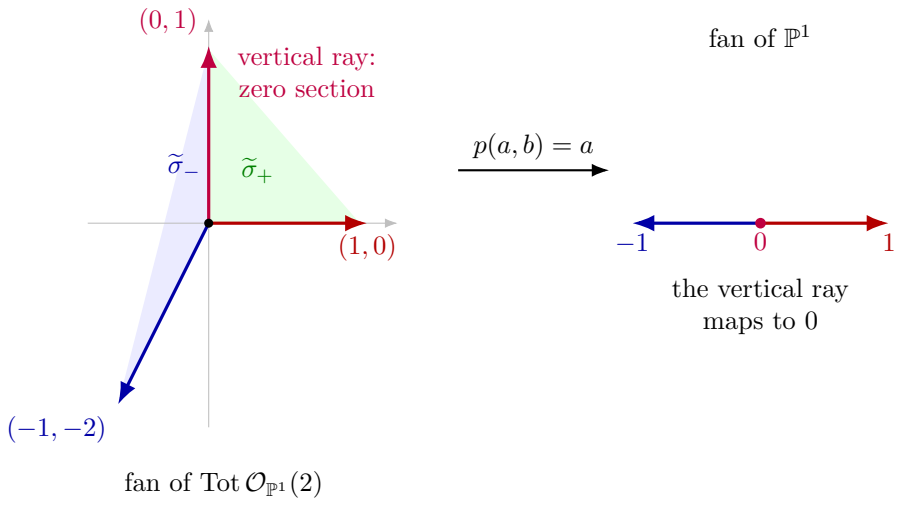


Diagram 3: The line-bundle fan in the convention of Chapter 6. For $D = 2D_{-1}$, the lifted rays are $(1,0)$ and $(-1,-2)$, together with the vertical ray $(0,1)$. Each maximal cone contains that vertical ray; projection onto the first coordinate is the bundle map.

5 Why remove the irrelevant locus? The fan explanation

The fan of $\mathbb{A}^r$ consists of all faces of the positive orthant. For $I \subseteq \Sigma(1)$, let

$$\tau_I = \mathrm{Cone}(e_\rho : \rho \in I).$$

A lattice map defines a toric morphism exactly when every source cone maps into a target cone. Thus we keep precisely the faces τ_I for which

$$\beta(\tau_I) \subseteq \sigma \quad \text{for some } \sigma \in \Sigma.$$

Since the u_ρ are rays of the same fan, this is equivalent to $I \subseteq \sigma(1)$. The resulting source subfan is

$$\widehat{\Sigma} = \{\tau_I : I \subseteq \sigma(1) \text{ for some } \sigma \in \Sigma\}.$$

Its toric variety is exactly $\widehat{X}_\Sigma$. It is the largest torus-invariant open subset of this $\mathbb{A}^r$ on which the given torus map extends regularly to X_Σ.

The orbit closure of τ_I in $\mathbb{A}^r$ is

$$V(\tau_I) = V(x_\rho : \rho \in I).$$

If I is forbidden, every larger set is forbidden as well, and its coordinate subspace is already contained in this one. Hence

$$Z(\Sigma) = \bigcup_{\substack{P \text{ minimal} \\ P \not\subseteq \sigma(1) \text{ for all } \sigma}} V(x_\rho : \rho \in P).$$

These minimal forbidden ray sets are the *primitive collections* in Chapter 6. This explains the minimal-set description of the irrelevant locus through the orbit–cone correspondence.

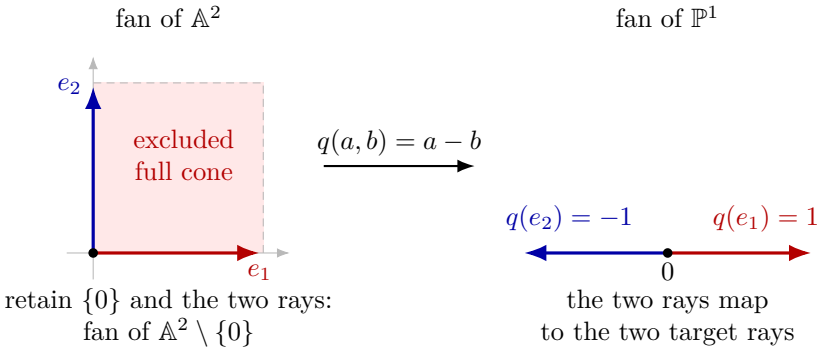


Diagram 4: The cocharacter map for the Cox quotient of $\mathbb{P}^1$. The full positive quadrant maps onto the whole real line, which is not contained in either cone of the target fan. Removing its orbit, the origin of $\mathbb{A}^2$, leaves a fan-compatible map.

Example 3 (Why the origin of $\mathbb{A}^2$ must be removed)**.** For $\mathbb{P}^1$, the lattice map is $\beta(a,b) = a - b$. The full positive quadrant maps to all of $\mathbb{R}$, which is not contained in either cone of the fan of $\mathbb{P}^1$. Algebraically, the map $[x : y]$ has no regular extension at the origin: both curves

$$t \longmapsto (t^2, t), \qquad t \longmapsto (t, t^2)$$

approach $(0,0)$, but their projective images approach $[0 : 1]$ and $[1 : 0]$, respectively. A morphism cannot give one point two different images. The torus map therefore defines a *rational map* with an indeterminacy at the origin. Arbitrary set-theoretic assignments at that point would not repair regularity. This is the precise meaning of the lecture's statement that the map is not defined there.

6 Quotients through invariant regular functions

Suppose G acts on a variety Y. A function on an orbit space must pull back to a function F satisfying

$$F(g \cdot y) = F(y) \qquad (g \in G).$$

Such a function is *invariant*. It is also equivariant if its target $\mathbb{A}^1$ is given the trivial action. Set-theoretically, invariance makes $f([y]) = F(y)$ well defined; an algebraic quotient must also have the correct regular functions and topology.

If $Y = \operatorname{Spec} A$ is affine and G is reductive over $\mathbb{C}$, the invariant algebra is finitely generated, and the affine quotient is

$$\boxed{Y /\!/ G = \operatorname{Spec}(A^G).}$$

The lecture's groups $(\mathbb{G}_m)^k$ and finite diagonalizable factors are included. Here A is the ring of *regular* functions, not merely the rational function field.

Why can this quotient identify distinct orbits? If F is constant with value c on an orbit, the closed set $F^{-1}(c)$ contains its closure. Thus two orbit closures meeting at one point have equal values under every invariant regular function. For an affine reductive action, each quotient fiber contains a unique closed orbit; the other orbits in that fiber have closures containing it.

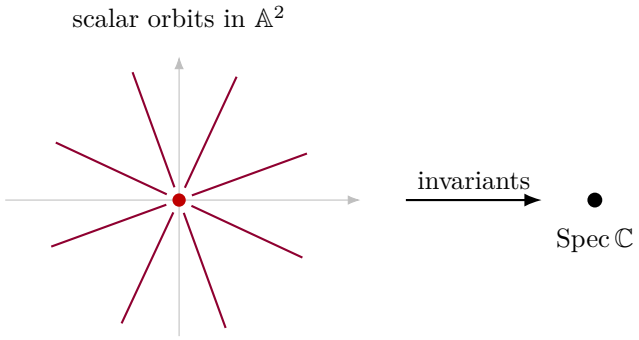


Diagram 5: Real slices of several complex scalar orbits. Each punctured line has the origin in its closure; the origin is a separate orbit. All invariant polynomial functions take the same value on all of them.

6.1 Good and geometric quotients

For the reductive actions in this lecture, a *good quotient* $\pi : Y \to Q$ is an invariant affine surjection with

$$\mathcal{O}_Q = (\pi_* \mathcal{O}_Y)^G,$$

which sends invariant closed subsets to closed subsets and sends disjoint invariant closed subsets to disjoint subsets. The sheaf condition means that regular functions on every open subset of Q are exactly the invariant regular functions on its inverse image. In our Cox construction this is checked by taking invariants on affine charts and gluing.

For a good Cox quotient, being *geometric* additionally means that every fiber is a single G-orbit. The Cox theorem says

$$\boxed{\widehat{X}_\Sigma \,/\!/\, G \text{ is geometric} \quad \Longleftrightarrow \quad \Sigma \text{ is simplicial.}}$$

This is why the double slash is useful before one has established the orbit statement. General definitions of geometric quotients also specify the quotient topology and invariant-function sheaf; geometric quotients in other settings need not be affine morphisms.

6.2 Recovering $\mathbb{P}^1$ by deleting, localizing, and gluing

Let $\mathbb{G}_m$ act on $\mathbb{A}^2$ by $t \cdot (x, y) = (tx, ty)$. A nonconstant monomial $x^a y^b$ has positive weight $a + b$, so

$$\mathbb{C}[x, y]^{\mathbb{G}_m} = \mathbb{C}, \qquad \mathbb{A}^2 \,/\!/\, \mathbb{G}_m = \operatorname{Spec} \mathbb{C}.$$

The nonzero vectors have many different orbits, but every orbit closure contains the origin. This affine quotient is a point, rather than $\mathbb{P}^1$.

Remove the origin. The open set $\mathbb{A}^2 \setminus \{0\}$ is not affine, so take a G-stable affine cover and compute locally:

open upstairs	coordinate ring	invariant ring
$D(x)$	$\mathbb{C}[x, x^{-1}, y]$	$\mathbb{C}[y/x] = \mathbb{C}[u]$
$D(y)$	$\mathbb{C}[x, y, y^{-1}]$	$\mathbb{C}[x/y] = \mathbb{C}[v]$
$D(xy)$	$\mathbb{C}[x^{\pm 1}, y^{\pm 1}]$	$\mathbb{C}[u, u^{-1}]$

The transition $v = u^{-1}$ glues the two quotient affine lines to $\mathbb{P}^1$. This is exactly the degree-zero localization construction: the grading records the action, and degree zero records invariance. The recording occasionally says $\mathbb{P}^2$ here; two homogeneous coordinates give $\mathbb{P}^1$.

Remark 3 (Global functions alone still do not recover the quotient). Even after deleting the origin,

$$\Gamma(\mathbb{A}^2 \setminus \{0\}, \mathcal{O}) = \mathbb{C}[x, x^{-1}, y] \cap \mathbb{C}[x, y, y^{-1}] = \mathbb{C}[x, y]$$

inside $\mathbb{C}(x, y)$. Its invariant subring is still $\mathbb{C}$. Thus taking Spec of the global invariant ring would again lose $\mathbb{P}^1$. The successful construction uses local invariant rings and their gluing. The same computation proves the punctured plane is nonaffine: its canonical map to the spectrum of its global functions is the inclusion into $\mathbb{A}^2$, which misses the origin.

For a general Cox chart, the corresponding identity is

$$\left(S_{\widehat{x^\sigma}}\right)^G = \left(S_{\widehat{x^\sigma}}\right)_0 \cong \mathbb{C}[\sigma^\vee \cap M].$$

This recovers Chapter 6's affine toric charts from the quotient.

6.3 The construction does not imply projectivity

The Cox quotient exists for fans that are not normal fans of polytopes. It therefore does not, by itself, prove projectivity or even quasiprojectivity. A toric variety is projective precisely when its fan is complete and admits an integral strictly convex support function. Every complete toric surface is projective; in dimension three, complete nonprojective examples occur. This is the distinction from Chapter 5 that arose in the lecture's question.

7 What changes for singular fans

The Cox construction uses the class-group sequence, not just Cartier divisors. Thus it continues to work when X_Σ is singular:

$$0 \longrightarrow M \longrightarrow \mathbb{Z}^{\Sigma(1)} \longrightarrow \mathrm{Cl}(X_\Sigma) \longrightarrow 0, \qquad G = \mathrm{Hom}(\mathrm{Cl}(X_\Sigma), \mathbb{G}_m).$$

As before, assume the rays span $N_\mathbb{R}$, so there is no torus factor. For a smooth variety $\mathrm{Pic}(X) = \mathrm{Cl}(X)$; for a singular variety, Weil-divisor classes need not represent line bundles.

If $\mathrm{Cl}(X) \cong \mathbb{Z}^k \oplus A_{\mathrm{tors}}$, then

$$G \cong \mathbb{G}_m^k \times \mathrm{Hom}(A_{\mathrm{tors}}, \mathbb{G}_m).$$

Over $\mathbb{C}$, the second factor is a finite diagonalizable group. Consequently, orbits *can* be disconnected. This does not say that every orbit is disconnected, nor that finite stabilizers require torsion in the global class group.

For a Cox point $z = (z_\rho)$, let

$$L(z) = \langle [D_\rho] : z_\rho \neq 0 \rangle \subseteq \mathrm{Cl}(X).$$

The nonzero coordinates constrain which rescalings can fix z, giving the formula

$$\boxed{G_z \cong \mathrm{Hom}(\mathrm{Cl}(X)/L(z), \mathbb{G}_m).}$$

At the distinguished point z_σ, whose coordinates are zero on $\sigma(1)$ and one elsewhere, this becomes

$$G_{z_\sigma} \cong \mathrm{Hom}(\mathrm{Cl}(U_\sigma), \mathbb{G}_m).$$

The *local* class group therefore controls the stabilizer over the orbit corresponding to σ.

7.1 A finite stabilizer with a connected Cox group

Label the displayed rays, in matrix order, by

$$u_1 = (1, 0), \qquad u_2 = (0, 1), \qquad u_3 = (-2, 1),$$

and take maximal cones $\sigma_1 = \mathrm{Cone}(u_1, u_2)$, $\sigma_2 = \mathrm{Cone}(u_2, u_3)$. The first cone is smooth, whereas $\det(u_2, u_3) = 2$, so the second is simplicial but singular.

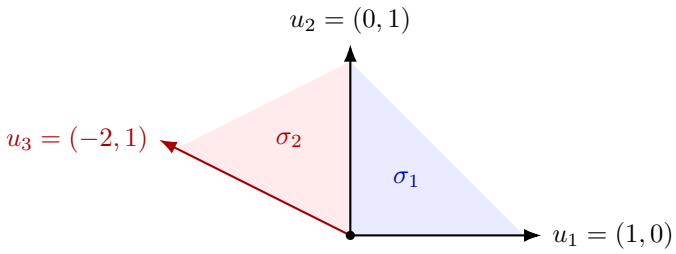


Diagram 6: The two-cone fan from the lecture. Its singular cone has index two, although the class group of the whole variety is free.

The character-divisor matrix and degree map are

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -2 & 1 \end{bmatrix}} \mathbb{Z}^3 \xrightarrow{(2,-1,1)} \mathbb{Z} \longrightarrow 0.$$

Indeed, $\ker(2, -1, 1) = \{(a, b, -2a + b) : a, b \in \mathbb{Z}\}$. Writing $H = [D_3]$, the relations are

$$[D_1] = 2H, \qquad [D_2] = -H, \qquad \mathrm{Cl}(X) = \mathbb{Z}H.$$

Thus the quotient data are

$$G = \mathbb{G}_m, \qquad \lambda \cdot (x_1, x_2, x_3) = (\lambda^2 x_1, \lambda^{-1} x_2, \lambda x_3),$$
$$B = (x_1, x_3), \qquad \widehat{X} = \mathbb{A}^3 \setminus V(x_1, x_3).$$

The point $z_{\sigma_2} = (1, 0, 0)$ is fixed exactly when $\lambda^2 = 1$, so its stabilizer is μ_2. Equivalently,

$$\mathrm{Cl}(U_{\sigma_2}) = \mathrm{Cl}(X)/\langle [D_1] \rangle \cong \mathbb{Z}/2\mathbb{Z}.$$

One can see the singularity directly on the chart $x_1 \neq 0$:

$$\mathbb{C}[x_1^{\pm 1}, x_2, x_3]^G = \mathbb{C}[x_1 x_2^2, x_2 x_3, x_3^2/x_1] \cong \mathbb{C}[A, B, C]/(AC - B^2).$$

Remark 4 (Correction to the displayed class group). For the rays in Diagram 6, the global class group is $\mathbb{Z}$, rather than the board's $\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z}$. The first two rows of the character matrix already have determinant one, ruling out torsion in its cokernel. The $\mathbb{Z}/2\mathbb{Z}$ is the *local* class group at the singular chart, and it correctly predicts the stated μ_2 stabilizer. This records the discrepancy for comparison with the forthcoming instructor handout without changing the pictured fan.

7.2 A nonsimplicial cone: several orbits in one fiber

The final example has ray generators

$$u_1 = (0,0,1), \quad u_2 = (1,0,1), \quad u_3 = (1,1,1), \quad u_4 = (0,1,1).$$

They span a three-dimensional cone over a square. Four rays in dimension three make it nonsimplicial.

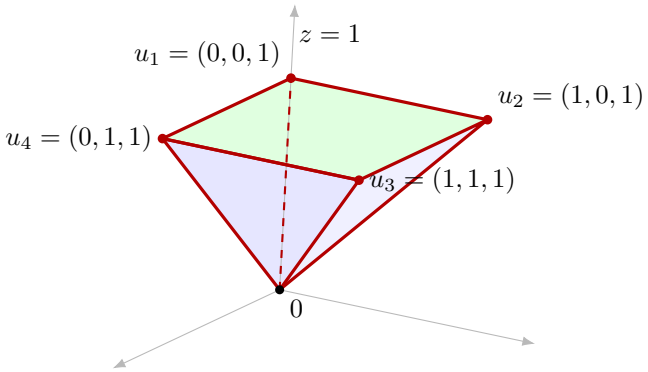


Diagram 7: The cone over the unit square in the plane $z = 1$. It has four extremal rays in a three-dimensional lattice, so it is not simplicial. The relation $u_1 - u_2 + u_3 - u_4 = 0$ gives the Cox weights $(1,-1,1,-1)$. The dashed ray is hidden behind the visible faces.

The relation $u_1 - u_2 + u_3 - u_4 = 0$ gives $\mathrm{Cl}(U_\sigma) \cong \mathbb{Z}$, with degrees $(1,-1,1,-1)$. Hence

$$\lambda \cdot (x_1, x_2, x_3, x_4) = (\lambda x_1, \lambda^{-1} x_2, \lambda x_3, \lambda^{-1} x_4).$$

There is one maximal cone containing every ray, so its complementary monomial is 1: $B(\Sigma) = (1)$ and nothing is deleted.

In the following display, B denotes an invariant coordinate, whereas $B(\Sigma)$ denotes the irrelevant ideal. The invariant ring is generated by four *monomials*

$$A = x_1 x_2, \qquad B = x_1 x_4, \qquad C = x_3 x_2, \qquad D = x_3 x_4,$$

with the single *binomial relation* $AD - BC = 0$. Therefore

$$\boxed{\mathbb{A}^4 /\!/ \mathbb{G}_m \cong \operatorname{Spec} \mathbb{C}[A,B,C,D]/(AD - BC).}$$

Every invariant monomial pairs equally many positive-weight and negative-weight variables, explaining these four generators.

The fiber over the vertex is

$$\pi^{-1}(0) = V(x_1, x_3) \cup V(x_2, x_4).$$

These are two coordinate planes *meeting at the origin*, not two disjoint orbits. In the punctured plane $x_2 = x_4 = 0$, the ratio $[x_1 : x_3] \in \mathbb{P}^1$ labels distinct scaling orbits. The other punctured plane gives another $\mathbb{P}^1$-family, and the origin is a separate fixed-point orbit. Thus the vertex fiber contains infinitely many orbits. All their closures contain the origin, so invariant regular functions cannot distinguish them.

Fan condition	Cox stabilizers	Quotient behavior
Smooth	Trivial; free G-action	Geometric: one orbit per fiber
Simplicial	Finite, possibly nontrivial	Still geometric
Nonsimplicial	Positive-dimensional at the relevant distinguished points	Good quotient; a fiber can contain several orbits

The last example distinguishes a quotient defined by invariant functions from a literal orbit space: the former may identify different orbits whose closures meet.

Further examples to discuss

1. Starting from the Cox weight matrix for $\mathbb{P}^1 \times \mathbb{P}^1$, add the fiber column for $\mathcal{O}(a, b)$ and check the chart transitions.
2. Repeat the singular surface calculation with the third ray $(-m, 1)$, $m \geq 2$. Locate the μ_m stabilizer and distinguish its local class group from the global one.
3. Compare the total-space fans of linearly equivalent divisors using the integral shear, and identify the zero section in each fan.

Connections and references

Chapter 6, Sections 6.3–6.4, supplies the irrelevant ideal and degree-zero localization construction. Section 6.6 supplies the stabilizer formula and the nonsimplicial quadric example. Its challenge exercise develops the additional-coordinate construction of line bundles used in this lecture.

For the general quotient language, see the Stacks Project entries Good quotients and Geometric quotients. The Cox quotient theorem is Cox–Little–Schenck, *Toric Varieties*, Theorem 5.1.11; see also David Cox's lectures on toric varieties.